Computer Science & IT



C programming

Structure & Union

One Shot



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Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Array of string

using array of character pointer

using 2D- array

Slide

Topics to be Covered



- Topic — Structure & union
- Topic — stdcat
- Topic — H.W problem
- Topic
- Topic

Slide

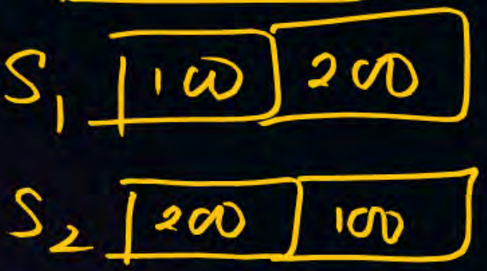


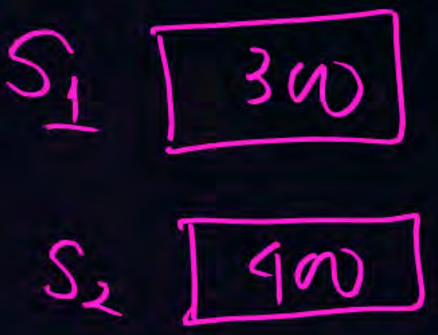
Question

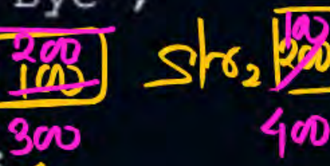
Consider the following C program:

```
#include<stdio.h>
void fun1(char *s1, char *s2){
        char *tmp;
        tmp = s1;
        s1 = s2;
        s2 = tmp;
}
void fun2(char **s1, char **s2){
        char *tmp;
        tmp = *s1;
        *s1 = *s2;
        *s2 = tmp;
}
int main (){
        char *str1 = "Hi",   *str2 = "Bye";
        fun1(str1,  str2);
        printf("%s %s ",  str1, str2);
        fun2(&str1,  &str2);
        printf("%s %s",  str1,  str2);
        return 0;
}
```









DS

Hi, Bye Bye Hi

temp: 100

*s1

(A) Hi Bye Bye Hi ✓

(B) Hi Bye Hi Bye

(C) Bye Hi Hi Bye

(D) Bye Hi Bye Hi

Slide



Question

What is printed by the following ANSI C program?

```
#include<stdio.h>
int main(int argc, char *argv[]){
   int x = 1, z[2] = {10, 11};
   int *p = NULL;
   p = &x;
   *p = 10;
   p = &z[1];
   *(&z[0] + 1) += 3;
   printf("%d, %d, %d\n", x, z[0], z[1]);
   return 0;
}
```

(A) 1, 10, 11

(B) 1, 10, 14

(C) 10, 14, 11

(D) 10, 10, 14



Question

What is printed by the following ANSI C program?

```
#include<stdio.h>
int main(int argc, char *argv[])
{
int a[3][3][3] = {{1, 2, 3, 4, 5, 6, 7, 8, 9},
                  {10, 11, 12, 13, 14, 15, 16, 17, 18},
                  19, 20, 21, 22, 23, 24, 25, 26, 27}};
int i = 0, j = 0, k = 0;
for( i = 0; i < 3; i++ ){
     for(k = 0; k < 3; k++ )
          printf("%d ", a[i][j][k]);
     printf("\n");
}
return 0;
}
```

(A) 1 2 3
10 11 12
19 20 21

(B) 1 4 7
10 13 16
19 22 25

(C) 1 2 3
4 5 6
7 8 9

(D) 1 2 3
13 14 15
25 26 27

Slide



Question

Consider the following program

```
#include <stdio.h>
int main(){

    int a[][2] = {12,34,5,6,78,8,99};
    int *ptr= *(a+1);

    printf ("%d", *ptr++);
    ++*ptr;
    printf ("%d", *ptr);
    return 0;
}
```

The output of the program is

Slide

a - Addrs 1-D

*a - Addrs int

**

100 104 108 116 124

112 120

5 7

5 7

57

ptr : *(100+1) = *(100+8)

ptr 108 112 = *108

Integer

Question

PYQ

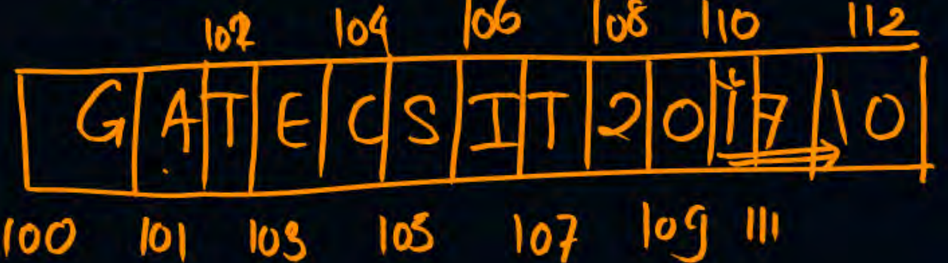
```
#Q Consider the following C Program.
#include <stdio.h>
#include< string.h>
int main () {
        char* c = "GATECSIT2017";
        char* p = c;
        printf("%d", (int) strlen (c+2[p]-6[p]-1));
        return 0;
}
```

The output of the program is _____.

17

10

2



100 + T - I - 1

2[p] = p[2]

100 + 20 - 9 - 1

17 strng

100 + 11 - 1 = 110

Slide



Question

Consider the following C program?

```
#include <stdio.h>
int a = 100;
void fun(int *x ){
      *x = *x+10;
      a = a-20;
}
int main(){
      a = a+50;
      fun(&a);
      printf("%d",a );
      return 0;
}
```

The value printed by above program is

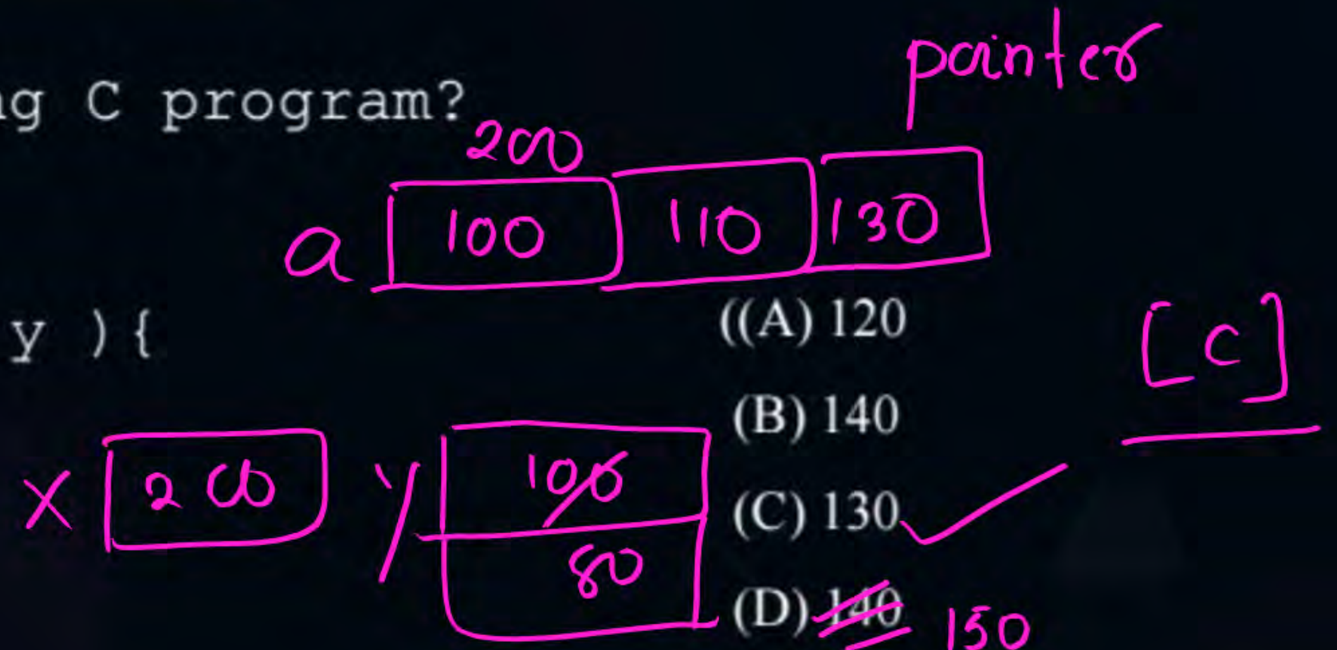
((A) 120
(B) 140
(C) 130
(D) 140

Question

Consider the following C program?

```
#include <stdio.h>
int a = 100;
void fun(int *x, int y ){
      *x = *x+10;
      y = y-20;
      a= a+20;
}
int main(){
      fun(&a,a);
      printf("%d",a);
      return 0;
}
```

The value printed by above program is



((A) 120

(B) 140

(C) 130

(D) ~~140~~ 150

[C]

#Q Consider the C program shown below.

```
#include <stdio.h>
int x;
void Q(int z) {
        z += x;
        printf("%d ", z);
}
void P(int *y) {
     int x = *y+2;
     Q(x);
     *y = x-1;
      printf("%d ", x);
     }
main(void) {
      x = 5;
      P(&x)
       printf("%d ", x);
}
```

The output of this program is

z = 7

7 + 5 = 12

z = z + x

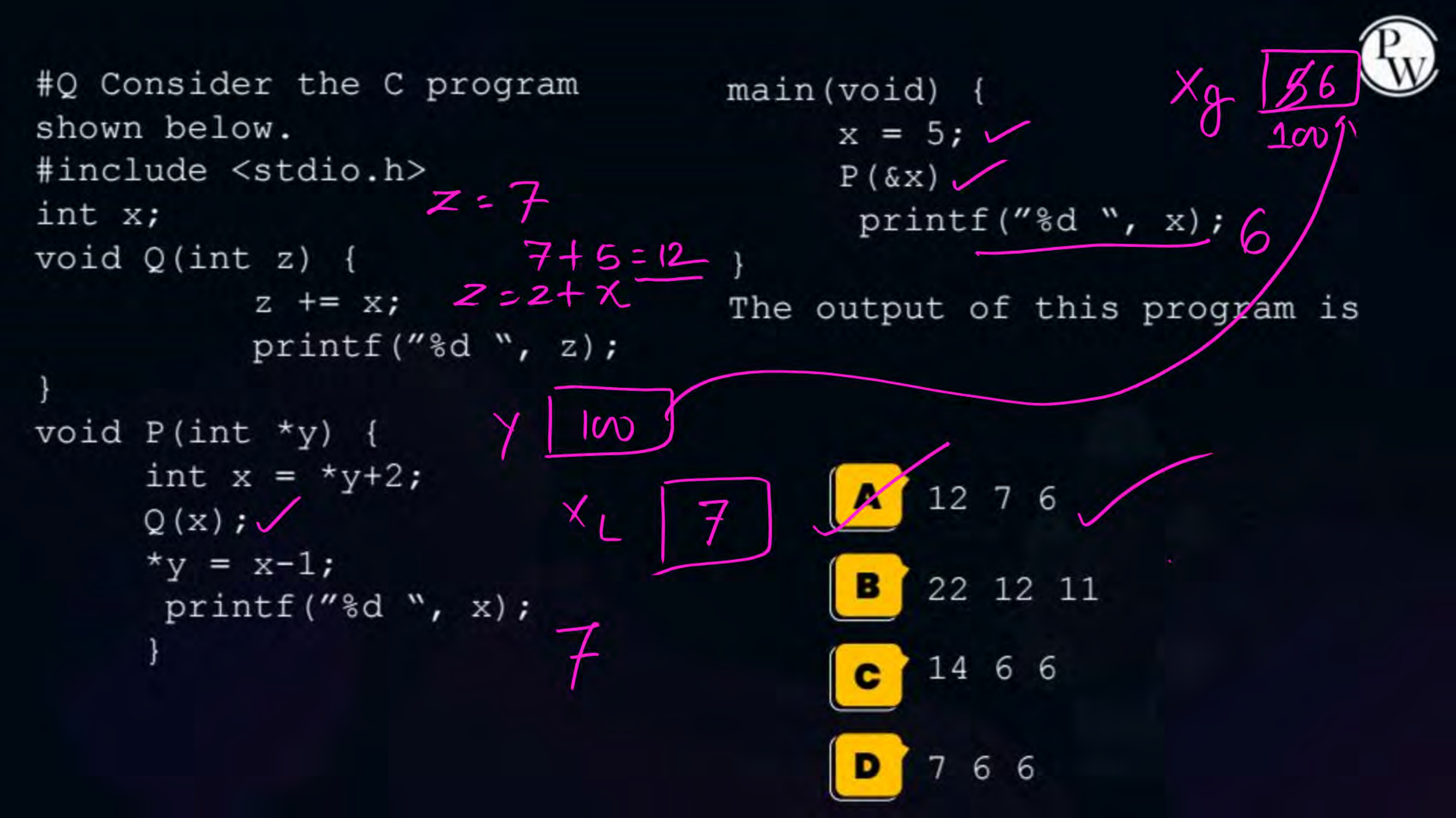
y | 100

x_L | 7

7

x_g | 56 | 100

6



(A) 12 7 6

(B) 22 12 11

(C) 14 6 6

(D) 7 6 6

Question

Consider the following C program?

```
#include <stdio.h>
void func(int *x,int *y,int *z){
    *y=*y+4;
    *z=*x+*y+*z;
}
int main()
{
    int x = 10, y=3;
    func(&y, &x, &x);
    printf("%d %d",x,y );
    return 0;
}
```

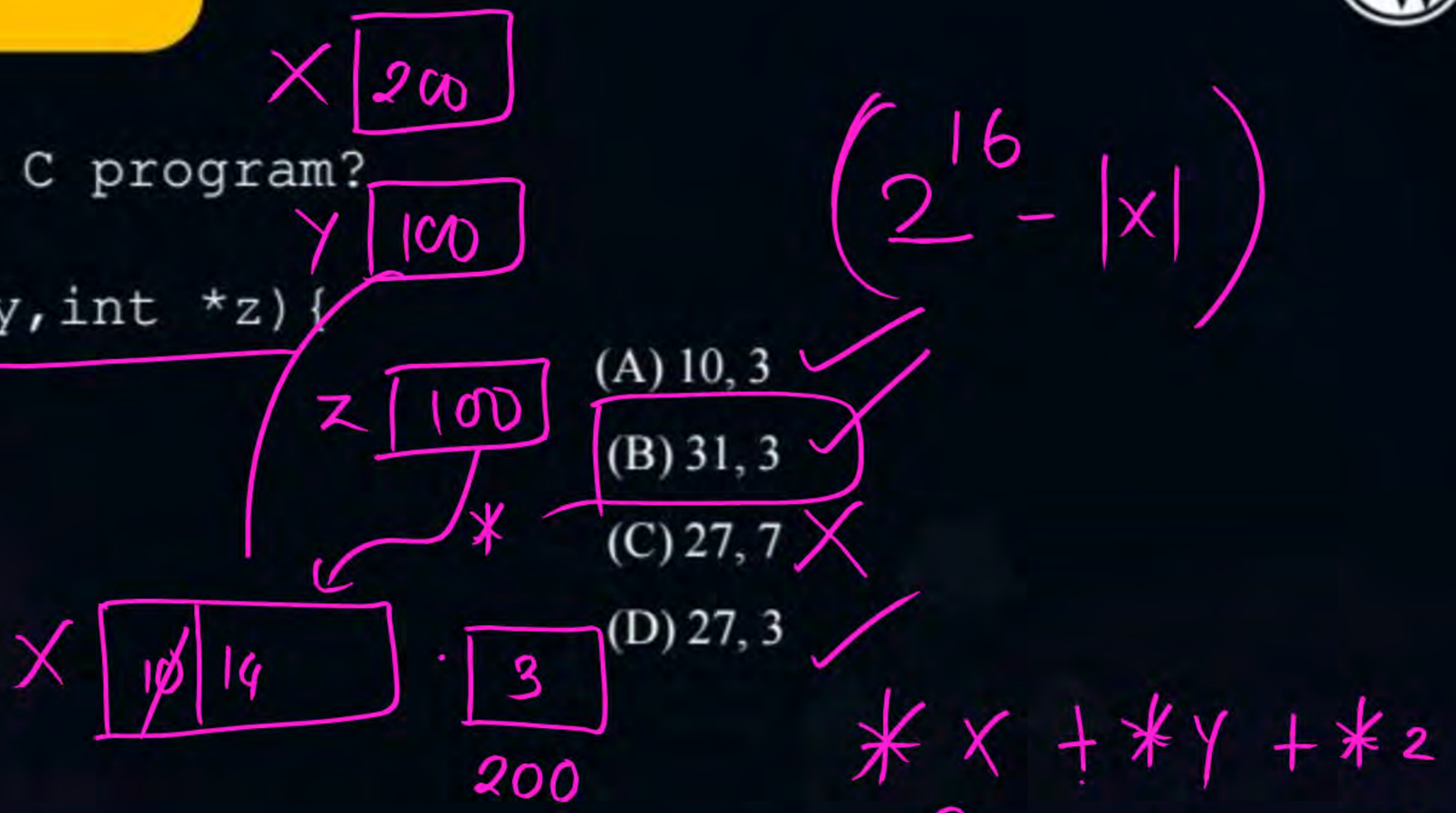
The value printed by above program is_

(A) 10, 3

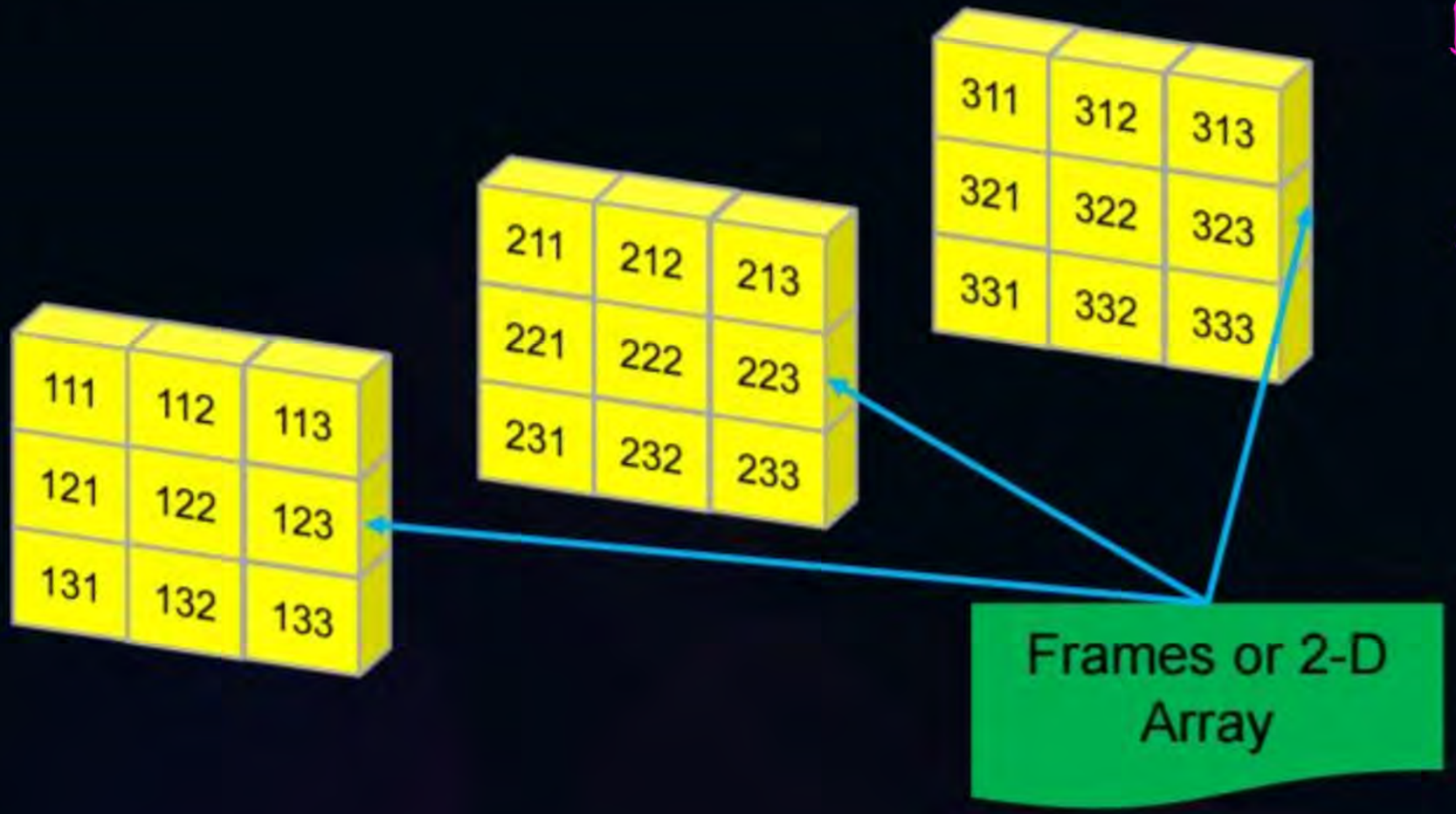
(B) 31, 3

(C) 27, 7

(D) 27, 3



3-D Array



C - Array & pointe

Data structur

Array :

Address

frame

Row

int a[3][3][3]

column

Slide



Structure

During declaration structure variable cant be initialize
Structure pointer
Arrow operator
Structure pointer
Typedef
Union



Structure

Structure is user defined data type

int a;

Structure is collection dissimilar data type

```
Struct test {
    int a;
    char ch;
};
```

Name of structure

Member of structure



Structure

Structure variable declaration

declaration

```
struct test t;
```

t is structure variable of type test

```
struct test t = {10, 'a'};
```

Initialization



Structure

To access member of structure we use '.' operator

```
t.a = 20;
t.ch = 'k';
printf("%d", t.a);
printf("%c", t.ch);
```

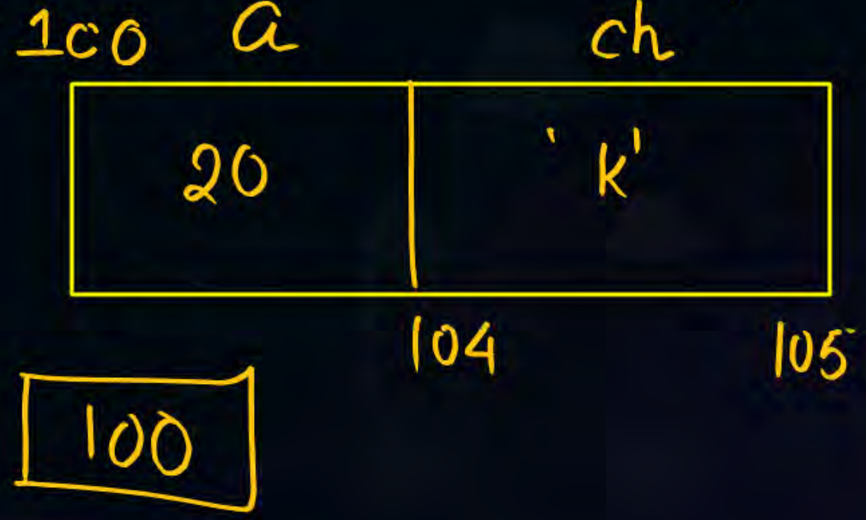
Slide

Structure

word alignment

```
Stouct test t;
Stouct test *pto;
pto = &t; // pto = 100
```

ptr is painter to structure of type test



Slide



Structure

→ dereference operator for structure pointer

```
Stouct test t;
Stouct test *pto;
pto = &t; //pto=100
pto → a = 30;
(*pto).a
pto→ch = 'L';
```

using structure pointer we can access Member of structure we use → operator

Slide



Structure

struct test $t_1[2]$: { {10, 'a'}, {11, 'b'} };

array variable

struct test *pto;

pto = t_1; ✓ //

Name of array itself Address

t_1++ ; // Error base Address is constant

pto++;

Slide



Union

Set Theory

union

OR

```
union test {
    int a;
    char ch;
};
```

max(a, ch)

```
union test t1;   <- variable
t1.a = 10;
t1.ch = 'a';

union test *ptr;
ptr = &t1;
ptr -> a = 20;
ptr -> ch = 'b';
```

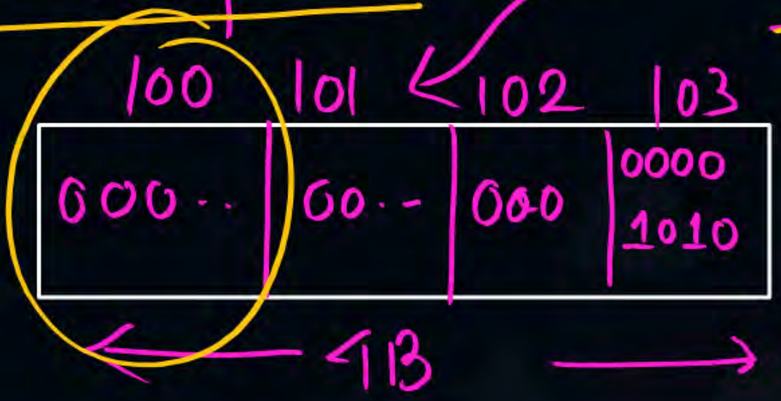
Slide

Union

In Union memory is allocated only for 1 Data type

The Data type with maximum space

"%d" - 4 Byte



"%c" - 1 Byte

Slide

Question

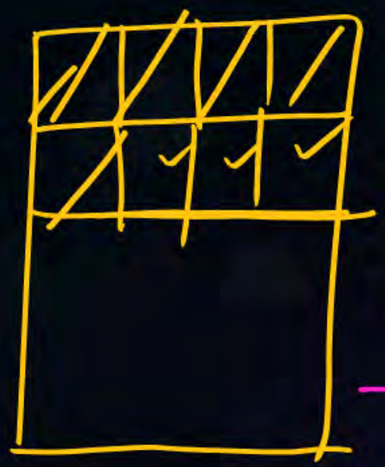
Q The following C declarations

```
struct node{
        int i:
        float j;
};
struct node *s[10];
```

define s to be

struct node s1[10];
Array of
Structure pointer

word 5Byte
word
Size : 4B



$\left\lceil \frac{5}{4} \right\rceil = \lceil 1.2 \rceil$

= 2 word

Address

(a) An array, each element of which is a ~~pointer~~ to a structure of type node

(b) A structure of 2 fields, each field being a pointer to an array of 10 elements

(c) A structure of 3 fields: an integer, a float, and an array of 10 elements

(d) An array, each element of which is a structure of type node

Slide

Question

Consider the following C declaration

```
struct {
        short s [5]
        union {
        float y;
        long z;
        } u;
}t;
```

Assume that objects of the type short, float and long occupy 2 bytes, 4 bytes and 8 bytes, respectively. The memory requirement for variable t, ignoring alignment considerations, is

(a) 22 bytes
(b) 14 bytes
(c) 18 bytes
(d) 10 bytes

struct
union
variable
2 4 8
struct test t1;

Addition Short
5x2 = 10 + 8 = 18
Max (float, long)
max (4, 8) = 8

struct test {
}

Slide

Question

[A] 0

Consider the following C program

```
#include<stdio.h>
struct Ournode{
char x,y,z;
};
int main(){
struct Ournode p = {'1','0','a' +2};
struct Ournode *q = &p;
printf("%c,  %c",   *((char*)q+1),   *((char*)q+2));
return 0;
}
```

The output of this program is:

(A) 0, c
(B) 0, a+2
(C) '0', 'a+2'
(D) '0', 'c'



97+2 = 99

1 if Not type casted then it will increement Size of structure

Slide

Strcat (char * dest, const char *src)

Concatenate : atlach source string at the end of destination

```
char ch1[20] = " parakram";
char ch[] = "vijay";
strcat(ch1, ch); ⟹ printf("%s", ch1);
```

parakramvijay



-32770

$\%$ +ve

$2^{16} - |x|$

$65534 - 32770$

$= 32766$



2 mins Summary

Topic

Topic

Topic

Topic

Topic

Structure

`.` `.` `→`

union

strcat

$C_n \times n!$

↑ Catalan No

Negative No. unsigned

X

$2^{16} - 1 \times$

32770

Slide

n - No. of keys

$$\frac{1}{n+1} {}^{2n}C_n$$

THANK - YOU

